

XRPL Validator — Daily Engineering Report • 2026-04-02

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **11 commits** (xrpl-validator 11)

Engineering changes

- **perf: parallel object fetching — 32 concurrent RPC requests per ledger** (2241766 , xrpl-validator) Replace sequential ledger_entry fetch loop with tokio semaphore-bounded parallel requests. 500 objects: 1.5s sequential → ~0.2s parallel.
- **perf: bump concurrent RPC fetches from 32 to 64** (4dd589f , xrpl-validator) Round time: 1.2-1.4s → 0.8-1.2s. Fetch overhead now ~0.1s. Bottleneck is pure hash computation (~1s), not network.
- **perf: depth-4 caching (65536 buckets) — hash compute 1s → 15ms** (ef3a806 , xrpl-validator) Replace 256 depth-2 buckets with 65536 depth-4 buckets. Only ~500 of 65536 buckets are dirty per ledger (0.8%) vs 230 of 256 (90%) before.
- **feat: Prometheus /metrics endpoint + stale cache fix** (02c7f15 , xrpl-validator)
- Rename /metrics HTML page to /dashboard - New /metrics returns Prometheus text exposition format (14 metrics) - Metrics: uptime, ledger_seq, consecutive_matches, total_matches/mismatches, round_time, compute_time, peers, state_objects, ready_to_sign, txs, info labels - Delete disk cache on invalidate_tree() to prevent stale cache scan loops - ValidatorMetricsSnapshot struct for clean data assembly
- **feat: SQLite history storage + /api/history endpoint** (931e666 , xrpl-validator)
- New history.rs module with HistoryStore backed by rusqlite (bundled SQLite) - Records every verified ledger round: seq, matched, healed, txs, objs, round_time_ms, compute_time_ms, peers, timestamp - /api/history endpoint returns 1h/24h/7d/30d summaries + recent entries - Wired into ws_sync: records after each push_sync_log call - Delete disk cache on invalidate_tree() to prevent stale cache scan loops - Growth: ...
- **feat: public validator page + incident log** (77a7fda , xrpl-validator)
- /validator: polished public-facing landing page with live metrics, uptime chart, hash verification status, architecture overview - /incidents: operational transparency page with historical incident timeline - Caddy config ready for validator.halcyon-names.io subdomain - Both pages pull from /api/state-hash and /api/history for live data - Clean corporate design with Inter font, responsive layout
- **feat: dashboard redesign, consistent nav, compute_time fix** (8a71356 , xrpl-validator)
- Restore original SSE-driven metrics page with smooth sparklines - Transaction Types moved to full-width bottom row with stable layout - All known tx types shown permanently (inactive dimmed at 30%) - Consistent nav across all pages: Home | Consensus | Peers | State | Metrics | GitHub - Root / serves public validator page - Old feed moved to /feed - Fix compute_time_secs not being stored in update_and_hash status - ...

- **fix: revert TLS cert/key mismatch from rejection to debug log** (`bddbb3f` , `xrpl-validator`) XRPL peers intentionally use ephemeral TLS keys separate from their node identity Public-Key. This is by design — authentication happens via session signatures, not TLS certificate pinning. The rejection was blocking all peer connections.
- **fix: metrics accuracy — engine hit rate shows -- when no txs applied, TLS check debug-only** (`f59d65b` , `xrpl-validator`)
- Engine hit rate: show "--" instead of "100%" when total_applied=0 - XRP Burned label: "All Time" → "Lifetime" for clarity - TLS cert/key mismatch: debug log only (XRPL uses separate TLS and protocol keys by design)

Docs and operations

- **security: fix critical and high audit findings** (`69835ca` , `xrpl-validator`)
- **security: fix all remaining audit findings (15 items)** (`bedb027` , `xrpl-validator`)

Appendix — commits

Time	Repo	Commit	Subject
17:18	<code>xrpl-validator</code>	<code>2241766</code>	perf: parallel object fetching — 32 concurrent RPC requests per ledger
17:43	<code>xrpl-validator</code>	<code>4dd589f</code>	perf: bump concurrent RPC fetches from 32 to 64
18:57	<code>xrpl-validator</code>	<code>ef3a806</code>	perf: depth-4 caching (65536 buckets) — hash compute 1s → 15ms
19:50	<code>xrpl-validator</code>	<code>02c7f15</code>	feat: Prometheus /metrics endpoint + stale cache fix
20:17	<code>xrpl-validator</code>	<code>931e666</code>	feat: SQLite history storage + /api/history endpoint
20:22	<code>xrpl-validator</code>	<code>77a7fda</code>	feat: public validator page + incident log
22:07	<code>xrpl-validator</code>	<code>8a71356</code>	feat: dashboard redesign, consistent nav, compute_time fix
22:23	<code>xrpl-validator</code>	<code>69835ca</code>	security: fix critical and high audit findings
22:33	<code>xrpl-validator</code>	<code>bedb027</code>	security: fix all remaining audit findings (15 items)
22:41	<code>xrpl-validator</code>	<code>bddbb3f</code>	fix: revert TLS cert/key mismatch from rejection to debug log
23:33	<code>xrpl-validator</code>	<code>f59d65b</code>	fix: metrics accuracy — engine hit rate shows -- when no txs applied, TLS check debug-only